

high returns since the early 1980s. Table 9.8 lists means, volatilities, and Sharpe ratios across maturities from January 1952 to December 2011 of excess bond returns, computing excess returns using the T-bill rate as the short rate. Like many researchers, I use the post-1951 sample because bond yields were pegged at artificially low yields by the Fed until the 1951 Treasury Accord between the U.S. Treasury and the Federal Reserve. This was a legacy of World War II when interest rates were held low so the government could borrow cheaply. The Treasury Accord freed the Fed from being forced to purchase bonds issued by the Treasury and gave the Fed independence.¹³ I split the sample at the end of 1982 because the Fed temporarily targeted monetary reserves in its conduct of monetary policy from October 6, 1979 to October 9, 1982.¹⁴

Table 9.8 shows that from 1952 to 1982, excess returns of bonds of five- to ten-year maturities were zero, on average, and excess returns of bonds with maturities greater than ten years were actually negative, at -0.62% . But since 1983, bonds have done very well as an asset class. Bonds with maturities of five to ten years and greater than ten years have returned 5.85% and 8.48% in excess of the T-bill rate from 1983 to 2011. Thus, the overall positive returns of these bonds of 1.91% and 2.97% , respectively, over the full sample are mostly due to the general decline in all interest rates post-1980. Over the full sample, bond returns increase, on average, with maturity: excess returns are 0.59% for bonds of less than one-year maturity and increase to 2.97% for bonds with maturities greater than ten years. Table 9.8 also separately breaks out 2000–2011, the Lost Decade for stock returns. (Stock returns are reported in the last column of Table 9.8.) Since 2000, bonds have been a terrific investment relative to stocks: bonds with maturities greater than ten years have returned 7.18% in excess of T-bills compared to -0.39% for U.S. equities.

The Sharpe ratios reported in Table 9.8 generally decrease across maturities starting from above 0.7 for bonds of maturities less than one year to around 0.3 for long-term bonds. Because average returns increase with maturities, the falling Sharpe ratios as maturities lengthen are due to the much greater volatility of long-term bond returns relative to short-term bond returns. This is in itself surprising—a ten-year bond should reflect the macro and financial environment over ten years, and given mean reversion of interest rates and the tendency of economic shocks to die out, this implies that long-term bonds should have significantly lower volatilities than short-term bonds. They don't. Volatilities of long-term bonds are only a little lower than short-term bonds. The large volatility of long-term bonds is indeed excessive compared to simple economic models

¹³ For an account of joint monetary and fiscal policy in the years after World War II, see Eichengreen and Garber (1991).

¹⁴ The Taylor rule is still appropriate over this sample, as Cook (1989) shows the short-term interest rate is a good instrument to proxy the Fed's actual monetary targeting operating procedure during this period.